RESEARCH ARTICLE

Cozie Singapore: Scalable crowdsourced smartwatch micro-surveys to capture longitudinal in-situ urban heat and noise perception

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ABSTRACT
The everyday experience of urban space is shaped as much by how people perceive heat, noise, and their social surroundings as by the physical conditions that sensors record. This paper characterizes how that lived experience differs across the common spaces of a dense tropical city using crowdsourced, watch-based micro-surveys collected with the open-source Cozie smartwatch platform. A field deployment in Singapore with 106 participants produced 9,962 geolocated micro-survey responses, each pairing self-reported thermal preference, noise, social context, and activity with the wearer's onboard physiological signals and with linked outdoor weather. Every response included a self-reported location in one of six everyday space types, and the reported experience was characterized through conditional perception fingerprints, relative distinctiveness expressed as lift, and effect-size-based association testing using the chi-square statistic, Cramér's V, and adjusted standardized residuals. The strongest separations between spaces are behavioral and acoustic rather than thermal, as what a person is doing and hearing identifies a space more sharply (Cramér's V up to 0.38) than how warm they feel (0.12). Commute spaces carry the loudest and most traffic-driven noise, indoor spaces are defined by talking, and cooling demand is lowest in air-conditioned offices and highest outdoors, while thermal preference tracks the immediate controlled environment more than the outdoor weather. These findings show that a scalable smartwatch micro-survey can recover and quantify the perceptual signature of everyday urban spaces, offering a human-centered evidence base for noise mitigation and for energy-conscious, thermal comfort-aware urban design in tropical cities.

KEYWORDS
Micro-ecological momentary assessments; Wearable sensing; Thermal preference; Urban noise perception; Human-centered urban design; Tropical cities

1. Introduction

A key component of urban studies is the characterization of how the built environment impacts humans. Often, this analysis uses conventional geospatial information from the data-rich mapping context. These geospatial contexts can then be enriched with a significant amount of metadata through convergence with temporal sensor data, imaging data from satellites or street-view imagery, or from smart devices. Street-view imagery has been used to quantify visual and perceived qualities of the streetscape

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at scale (Ito, Kang, Zhang, Zhang, & Biljecki, 2024), while network-based methods characterize the morphology and connectivity of the urban fabric (Yap, Stouffs, & Biljecki, 2023). Even human presence itself can be inferred indirectly, for example by estimating building occupancy from mobile phone traces (Barbour et al., 2019).

Despite this effort, significant elements of the human experience in the urban context are missing as observation is not the equivalent of capturing perception. Physical measurements alone are weak predictors of how occupants actually judge their surroundings, because non-physical and psychological factors shape perception (Castaldo et al., 2018), and expectations condition the same measured conditions differently (Schweiker, Rissetto, & Wagner, 2020). Even the most widely used comfort model predicts individual thermal sensation with limited accuracy across large field datasets (Cheung, Schiavon, Parkinson, Li, & Brager, 2019), underscoring the gap between what sensors record and how conditions are judged. Image-based surveys offer one way to elicit perception at scale, asking people to rate qualities such as safety, liveliness, and greenery from street-view imagery (Gu et al., 2025; Ito et al., 2024). However, this perception is elicited from viewing scenes remotely rather than from lived, in-situ experience, so it misses characterizing physical responses from senses beyond sight, as well as environmental and social factors (Quintana et al., 2025). Capturing perception directly has a long methodological basis in ecological momentary assessment and experience sampling, which record subjective states in real time and in context (Christensen, Barrett, Bliss-Moreau, & Lebo, 2003; Stone & Shiffman, 1994), and pairing these reports with location data extends them to the geographically explicit case (Kirchner & Shiffman, 2016).

1.1. Crowdsourcing humans-as-a-sensor data across the urban context

Geographically-explicit ecological momentary assessment (GEMA) developed as a set of methods to record momentary subjective states together with the location in which they occur, and it has been applied across public health and urban analysis (Kirchner & Shiffman, 2016). It has been used to link the daily acoustic environment to real-time noise annoyance (Zhang, Zhou, Kwan, Su, & Lu, 2020) and to relate exposure to natural diversity to momentary mental wellbeing (Hammoud et al., 2024). Standardized architectures and reporting guidelines have since emerged to make these studies more comparable and reproducible (Gharani et al., 2021; Kingsbury et al., 2024). Wearables are increasingly used to capture indoor and environmental quality alongside the people experiencing it (Abboushi et al., 2022), and consumer smartwatches can measure exposures such as ambient noise with useful accuracy (Fischer, Schraivogel, Caversaccio, & Wimmer, 2022). Citizen-science deployments have likewise shown that non-experts equipped with consumer devices can map urban noise pollution at scale (Zipf, Primack, & Rothendler, 2020). The open-source Cozie platform takes this approach a step further by delivering micro-surveys directly on a smartwatch and pairing each response with the device's onboard physiological and environmental sensors (Tartarini, Frei, Schiavon, Chua, & Miller, 2023). The Cozie-based micro-survey approach has recently been applied to characterize sleep habits and bedroom environments (Miller, Chua, Frei, et al., 2025).

This study was developed as an effort to scale up this type of data collection, with a focus on heat and noise in the urban context. Both exposures are central to the everyday experience of a dense tropical city, where thermal comfort is a persistent concern and noise is a leading source of distraction and dissatisfaction (Appel-Meulenbroek, Steps, Wenmaekers, & Arentze, 2020). Individuals also differ substantially in how they perceive the same conditions, which has driven a shift toward personal comfort models built at the level of the person rather than the building, since people differ in their intrinsic thermal sensitivity (Arakawa Martins, Soebarto, & Williamson, 2022; Rupp, Parkinson, Kim, Toftum, & de Dear, 2022; Wang et al., 2018). Linking these reports to geolocated urban context connects them to the digital-twin models increasingly used to represent cities (Ignatius et al., 2024). The resulting dataset previously supported a public machine-learning competition on predicting heat and noise perception from contextual data (Miller, Ibrahim, et al., 2025; Miller et al., 2023), and this paper instead characterizes the collected experience itself across the range of everyday urban spaces.

2. Methods

This study characterizes human-experience data collected in Singapore using the Cozie platform, an open-source application for the Apple Watch that administers watch-based micro-surveys and records the wearer's onboard physiological and environmental data from the sensors (Tartarini et al., 2023). The overall approach was first proposed as a smartwatch-driven just-in-time intervention framework for building occupants (Miller, Chua, Frei, & Quintana, 2022), and the deployment and its instrumentation have since been described in methods papers that also report preliminary results (Miller, Ibrahim, et al., 2025; Miller et al., 2023), with the same platform underpinning related work on just-in-time adaptive interventions (JITAI) for heat and noise (Miller, Chua, Quintana, et al., 2025). The design follows many of the established practices for environmental surveys and ecological momentary assessment and its geographically explicit extension, including recommendations for sampling, scheduling, and transparent reporting (Graham, Parkinson, & Schiavon, 2021; Kingsbury et al., 2024; Liao, Skelton, Dunton, & Bruening, 2016; Reichert et al., 2020). Because the platform continuously records location and physiological signals, the deployment also considers the issues raised for GPS- and mobile-sensing studies (Fuller, Shareck, & Stanley, 2017; Kerr, Duncan, & Schipperijn, 2011). The remainder of this section describes the field study and participants, the structure of the micro-survey, the linked wearable, weather, and urban-context data, and the processing steps that produce the survey-level dataset used for the characterization that follows.

2.1. Crowdsourced watch-based micro-survey and physiological and environmental data

The field study methodology reported here is the same deployment described in detail by Miller, Chua, Quintana, et al. (2025), and the following is a summary of that protocol. The key elements of the protocol that are used in this analysis are illustrated in Figure 1. Participants were recruited through the National University of Singapore (NUS) Student Work Scheme and by word of mouth, producing a pool made up mostly of NUS students together with a smaller number of adults from outside the university. The cohort was predominantly young, with roughly 73% aged 18 to 30, and comprised 39% male and 61% female participants, and ethical approval was granted by the NUS Institutional Review Board. Each participant used an Apple Watch running the Cozie application paired with an iPhone, and loan devices were provided to those who lacked a compatible smartwatch. Participants were asked to wear the watch during working hours, at least from 9 AM to 7 PM on weekdays, for approximately four weeks. They were instructed to complete 100 micro-surveys together with weekly surveys and an in-person exit interview. Some participants provided responses outside the parameters in these guidelines, and those submitted within Singapore were retained in this analysis.

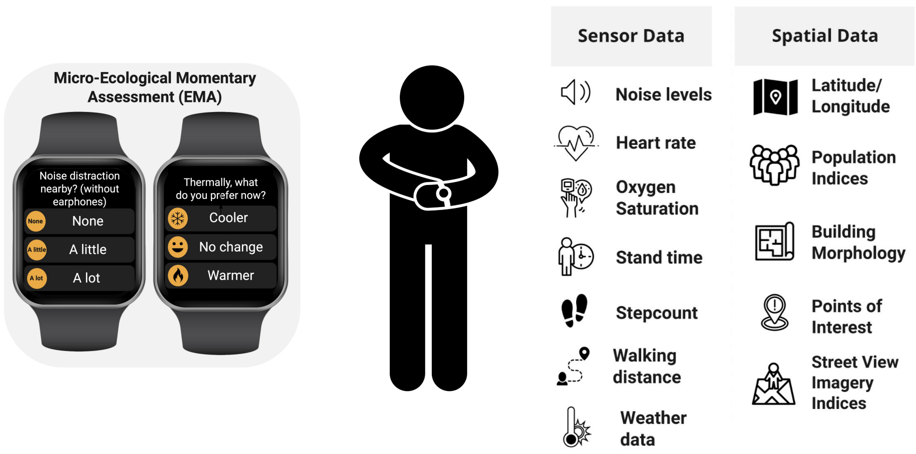

Figure 1. Overview of study design linking crowdsourced smartwatch micro-surveys (simple examples of questions on left) to the wearable sensor, weather, and urban spatial data each response is joined to (right) (Adapted from (Miller et al., 2022)).

As shown in Figure 1, each watch-based micro-survey response is time-stamped and geolocated, and is paired with the wearer's onboard physiological signals and with linked outdoor weather and urban-context data. The present analysis draws on a slightly different prepared dataset from this deployment, comprising 9,962 micro-survey responses from 106 participants, and focuses on characterizing the reported experience rather than the intervention outcomes.

Figure 2 illustrates the set of micro-survey questions answered directly on the watch face at each full submission instance. A response records the participant's thermal preference (*Cooler*, *No change*, or *Warmer*) and the level of nearby noise (*None*, *A little*, or *A lot*). When noise is present, follow-up questions capture the kind of noise and whether earphones are being worn. The survey also records the reported location or space type, whether the participant is *Alone* or in a *Group*, and the activity being performed, with the activity options depending on whether the participant is alone or with others. Several of these questions are conditional, so later items, such as noise kind and activity, are answered only for responses that reach them.

Each geolocated response was linked to a set of urban-context indicators derived with Urbanity, an open-source tool that builds feature-rich urban networks from open geospatial data (Yap et al., 2023). These indicators summarize the built environment around each response location, including streetscape view indices for greenery, sky, buildings, and roads, as well as visual complexity, building count, population, and nearby points of interest across categories such as commercial, food, and recreational uses. They provide the objective urban characterization used to test whether physical context adds information beyond the self-reported experience.

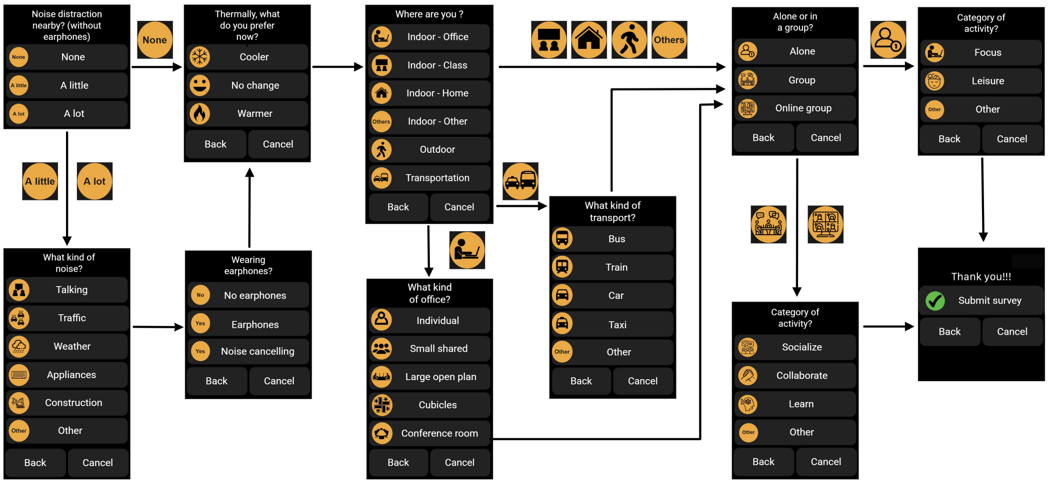

Figure 2. Full smartwatch micro-survey structure showing the left-to-right flow of watch screens from the core questions and their conditional follow-ups to submission (Adapted from (Miller et al., 2022))

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2.2. *Data processing and characterization*

The raw deployment data combines high-frequency watch and health streams with the sparser watch-survey responses in a single time-indexed record. Survey responses were first screened to those triggered on the watch that carried valid, non-zero coordinates falling inside the Singapore bounding box, which produced the 9,962-response analysis set. For each retained response, the seven temporal watch-health streams were aggregated over three pre-survey windows of six hours, one hour, and ten minutes, summarizing each window by its count, mean, spread, and extremes, and by an accumulated total for step count, walking distance, and stand time. The nearest weather-station observations were aggregated over the same three windows, and a heat index was derived from air temperature and relative humidity. These health and weather aggregates were then merged back onto the survey rows and checked for consistency, yielding one analysis-ready record per response that includes the original survey and location fields, along with the linked physiological, weather, and urban-context features used throughout the analysis.

The reported experience was then characterized against the reported space type using the analysis-ready record described above. Each response was categorized into one of the space types based on that survey response, and each categorical perception variable was cross-tabulated with the space type to form the contingency tables used below. Conditional questions, such as noise type and earphone use, have lower coverage, so their profiles describe only the responses for which follow-up was observed. Because responses are clustered within participants, the significance markers reported below are treated as descriptive screens, and effect sizes and the coherence of contrasts across related variables are used as the primary read-outs. The three methods below produce the perceptual characterization used for analysis.

2.2.1. *Conditional perception fingerprints*

For each space type s and response category r, the conditional proportion is the maximum-likelihood estimate of the categorical distribution within that space,

$$\hat{P}(\text{response} = r \mid \text{space} = s) = \frac{n_{rs}}{n_s}, \tag{1}$$

where n_{rs} is the number of responses of category r in space s and n_s is the total number of responses in that space. Each row, therefore, describes a single space's response profile and sums to 1 within each survey variable. No smoothing or shrinkage is applied, so cells based on few responses are noisier than their shading suggests.

2.2.2. *Relative distinctiveness through lift*

To separate what is distinctive to a space from what is simply common everywhere, each conditional proportion is divided by the corresponding corpus-wide proportion to give the lift,

$$\text{lift}(r, s) = \frac{P(\text{response} = r \mid \text{space} = s)}{P(\text{response} = r)}. \tag{2}$$

A value of one indicates a response that is as common in the space as in the corpus overall, values above one indicate over-representation, and values below one indicate under-representation.

2.2.3. *Association strength and cell-level contrasts*

The overall association between space type and each perception was tested with a Pearson chi-square statistic on the $r \times k$ contingency table,

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \tag{3}$$

where O_{ij} and E_{ij} are the observed and expected counts under independence. Because χ^2 grows with sample size, it was rescaled to a bounded effect size using Cramér's V,

$$V = \sqrt{\frac{\chi^2}{n\,(\min(r, k) - 1)}}, \tag{4}$$

interpreted with conventional descriptive bands (negligible < 0.1, weak 0.1–0.2, moderate 0.2–0.3, strong > 0.3). To identify which individual space and response combinations drive the association, adjusted standardized residuals were computed for each cell,

$$z_{ij} = \frac{O_{ij} - E_{ij}}{\sqrt{E_{ij}\,(1 - r_i/n)\,(1 - c_j/n)}}, \tag{5}$$

where r_i and c_j are the row and column totals and n is the grand total. Under independence, each z_{ij} is asymptotically standard normal, so cells with $|z| > 1.96$, 2.58, and 3.29 exceed the conventional 0.05, 0.01, and 0.001 thresholds and mark the space and response pairs shown as over- or under-represented.

3. Results

The collected data support an analysis focused on the volume, coverage, and timing of the responses, as well as the distributions of the survey and its linked physiological and weather measurements. The analysis then focused on how the reported experience differs across the everyday spaces in which it was recorded.

3.1. Data collection, coverage, and temporal characteristics

Figure 3 shows the mapping of the responses across Singapore. The densest grid cells sit over residential estates and the NUS university campus, consistent with the roughly 45% of responses reported from home, while coverage thins quickly toward the periphery even as individual responses still reach the central business district and points along the transport network.

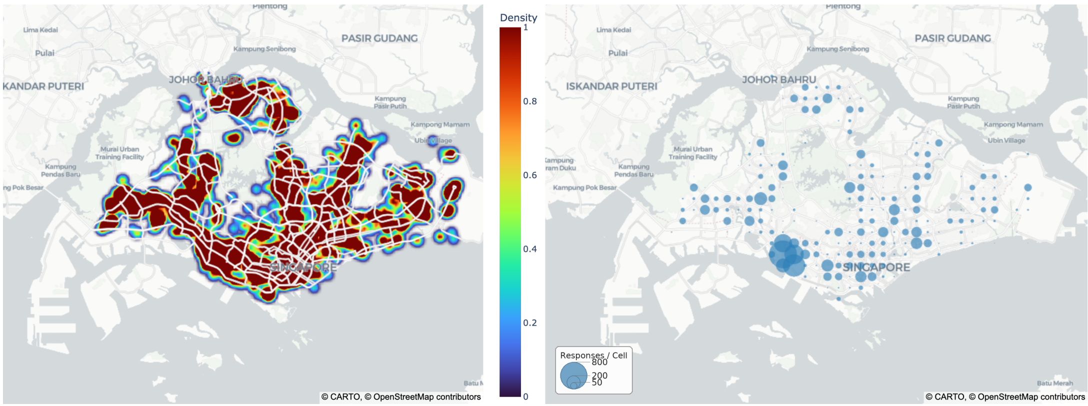

Figure 3. Spatial coverage of survey responses across Singapore, shown as a kernel-density surface of response locations (N = 9,931; left) and response counts aggregated into approximately 1-km grid cells (276 cells, top cell 897 responses; right).

Figure 4 illustrates the breakdown of data collection across the cohort of 106 participants. Contribution is spread evenly enough that no small group dominates the dataset, with a median of 98 responses per person, close to the 100-survey target, and the most active tenth of participants supplying only about 12% of the total. The range from 17 to 158 responses does mean that some participants are somewhat unequally represented, but the analyses that follow rely on within-space proportions and effect sizes rather than raw response counts.

The temporal cadence in Figure 5 reflects a protocol built around the working day. Responses concentrate sharply between 9 AM and 7 PM and peak in the mid-afternoon, and weekdays account for all but about 17% of the data, so the dataset captures waking, active hours more than nights or weekends. The calendar-week panel shows the response volume rising and falling with the phased recruitment of participants.

Timing is normalized within each space type in Figure 6, so it reads as a set of temporal signatures for when data is provided in that category. Offices and classrooms are almost exclusively weekday, daytime settings, with about 96% of office responses

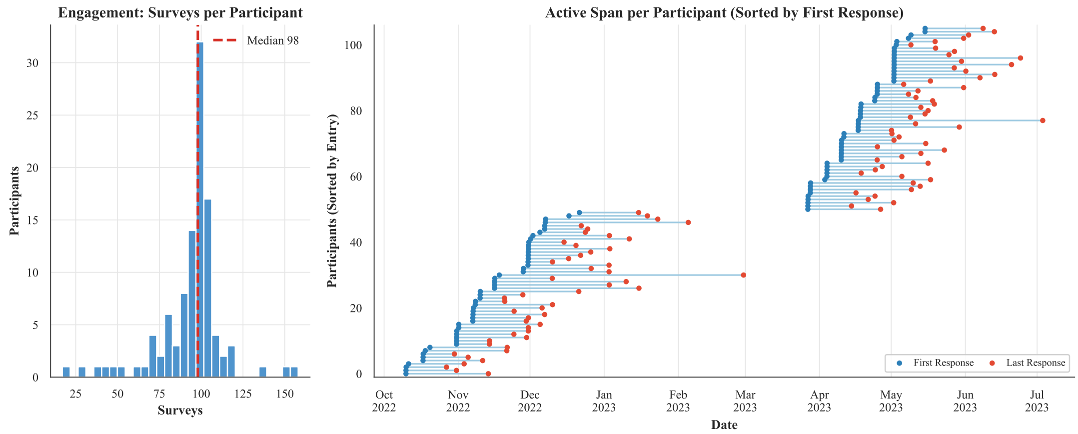

Figure 4. Participant engagement over the deployment: the distribution of surveys per participant (left; median 98) and each participant's active span from first to last response (right).

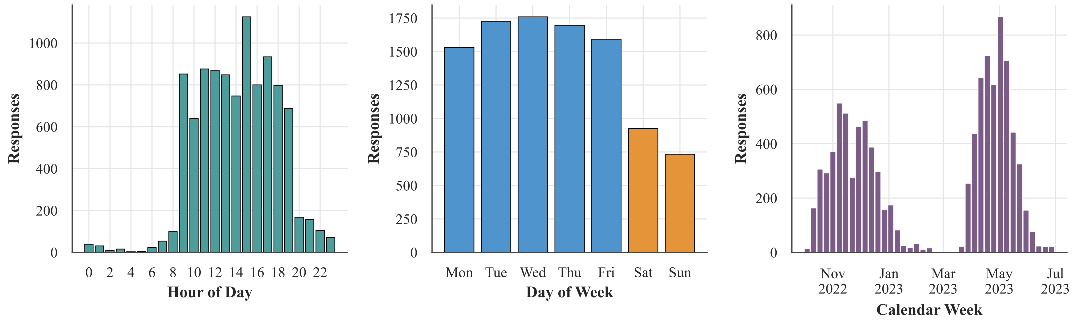

Figure 5. Time frames of survey responses in Singapore local time by hour of day (left), day of week (middle), and calendar week (right).

on weekdays, while home stretches into the evening and holds the largest weekend share of any space. The commute environments have a unique fingerprint, as outdoor and transportation responses have peakes in the morning and early evening following commute patterns.

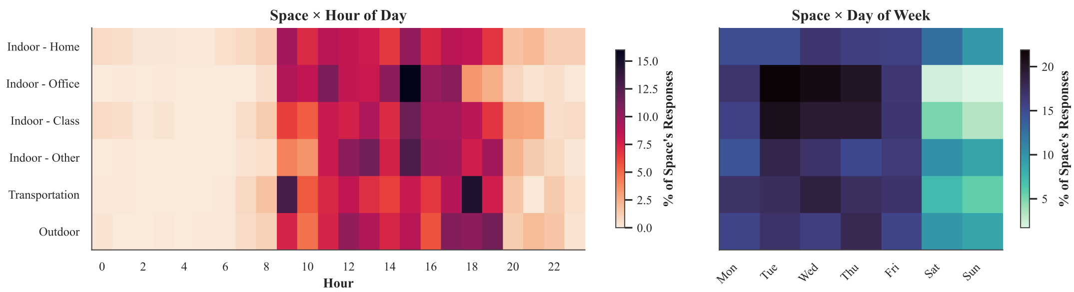

Figure 6. Row-normalized response timing by space type, showing each space's share of responses across hour of day (left) and day of week (right).

3.2. Survey responses and linked temporal data coverage

Figure 7 gives the overall distribution of every question and these data provide the breakdown of the participant self-assessments. Thermal preference is asymmetric, with a small majority wanting *No change* (54%), with much of the rest wanting to be *Cooler* (40%), and *Warmer* more rare (6%). This pattern is expected for a warm, humid climate, though it also reflects that many indoor spaces are air-conditioned rather than exposed to the outdoor weather. Perceived noise distraction is more balanced, with roughly equal shares reporting *None* (46%) and *A little* (44%) and only about one in ten reporting *A lot*. Noticing some noise is the norm, while a strong distraction is uncommon. Where noise is present, the follow-up question most often attributes it to people talking (44% of noise-type responses), well ahead of appliances and traffic. Most responses come from home (45%), participants are usually *Alone* (72%) rather than in a *Group*, and the activity questions show focused work and leisure in solitary moments and socializing in group ones. Earphone (or audio headphone) use is uncommon, even though the question is included in about half of surveys, as most responses report *No earphones*, and active noise canceling is rare. Therefore, participants seem to tolerate rather than mask their sound environment. The coverage panel shows that linked analysis is feasible with the one-hour pre-survey window, including weather for about 96% of surveys and heart rate, movement, and measured noise for roughly 90%, so nearly every self-report has matching sensor data.

The same responses appear as a flow through the questions in Figure 8, which makes the conditional structure clearer through a sankey diagram. Reading from left to right, each answer splits into the next question, the conditional branches, such as noise kind and earphones, carry only half of the responses that reach them, and activity is divided by whether the participant is *Alone* or in a *Group*. The ribbon widths are drawn to scale, so the proportion of every category shows that the common categories such as *No change* or *Cooler* are more prominent.

The linked wearable health streams are shown in Figure 9 with each parameter summarized over the six-hour, one-hour, and ten-minute windows preceding a survey. Median step count and walking distance fall as the window narrows, from about 85 to 42 steps per reading between the six-hour and ten-minute windows, because a short look-back captures only the last few minutes of activity. Heart rate is more stable at roughly 80 bpm across all three windows. Coverage moves the other way and limits how tight the window can be. The six-hour window carries movement and noise for over 95% of surveys, but at ten minutes, audio coverage falls to about a third, and step coverage to little more than half. Resting heart rate and blood-oxygen streams are sparse in every window. The one-hour window is used as the aggregate method throughout the rest of this analysis, since it maintains coverage near 90% while staying close to the survey moment.

Figure 10 summarizes the nearest-station outdoor weather over the same windows, and its distributions are narrow and stable, as expected for an equatorial climate. Air temperature has a median near 29 °C and relative humidity near 73%, which gives a heat index around 34 °C that changes little across the windows. The outdoor thermal environment is consistently warm, whatever the window length. Rainfall is the one strongly window-dependent feature because it accumulates over time, with about 28% of six-hour windows containing some rain against only 5% of ten-minute windows. These values describe outdoor conditions at a station, a median of roughly 3 km, from the response. It should be noted that these data are not the participant's actual microclimate, which is often indoor and air-conditioned.

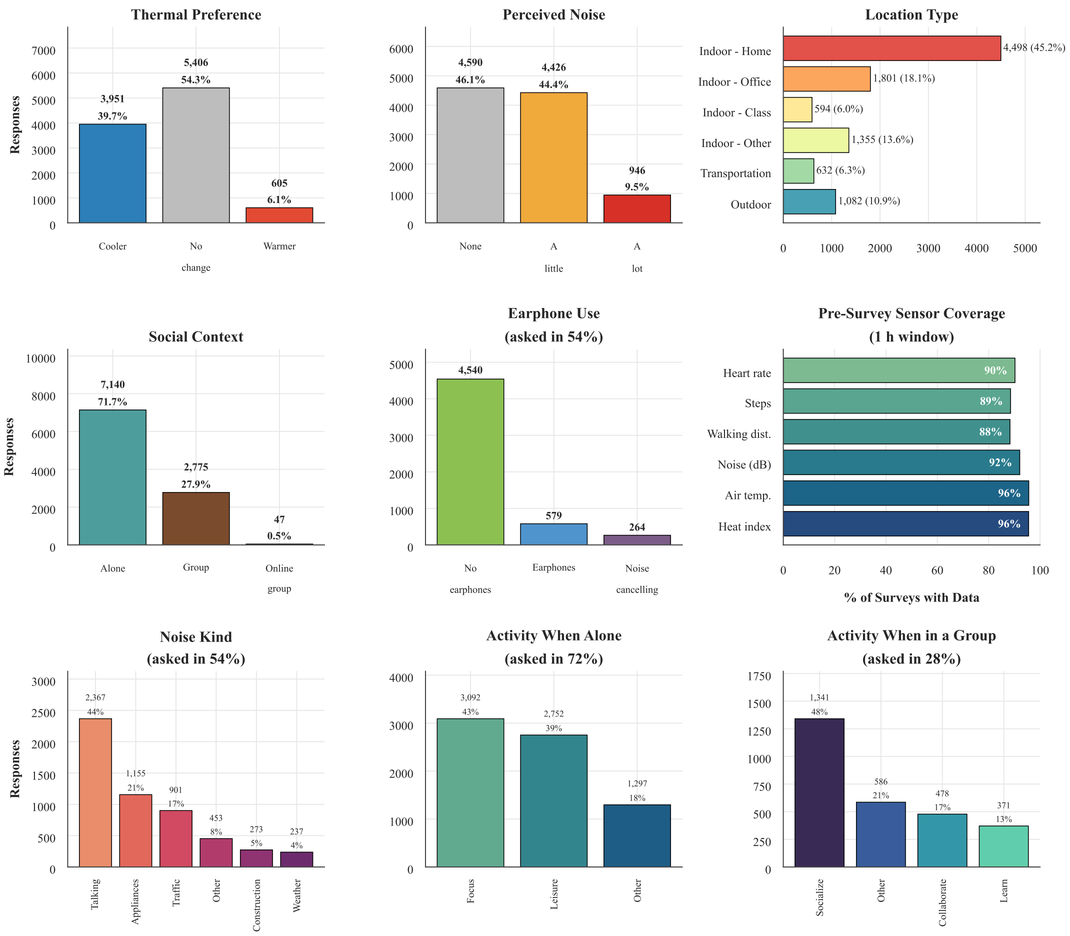

Figure 7. Response distributions for each survey item and pre-survey linked-sensor coverage (N = 9,962 responses, 106 participants).

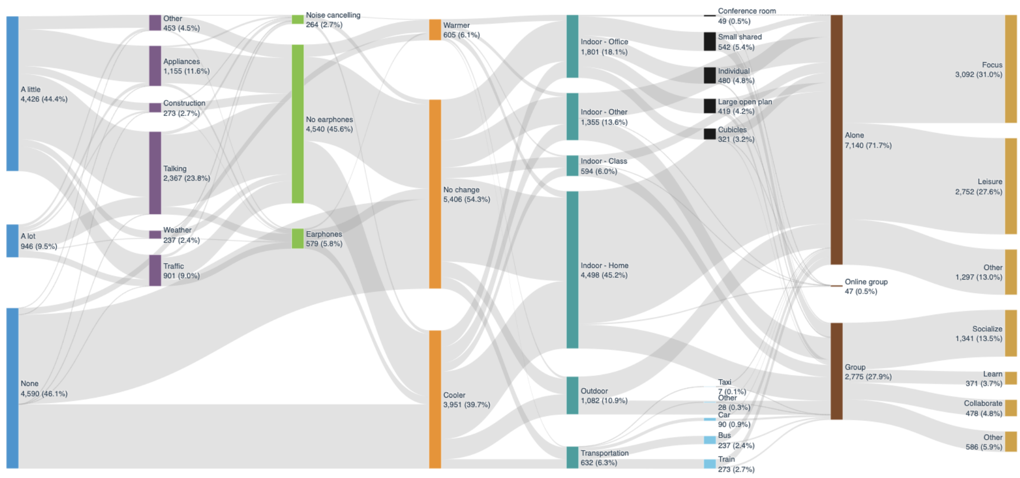

Figure 8. Sankey diagram of the conditional response flow through the questionnaire, with band width proportional to the number of responses along each path.

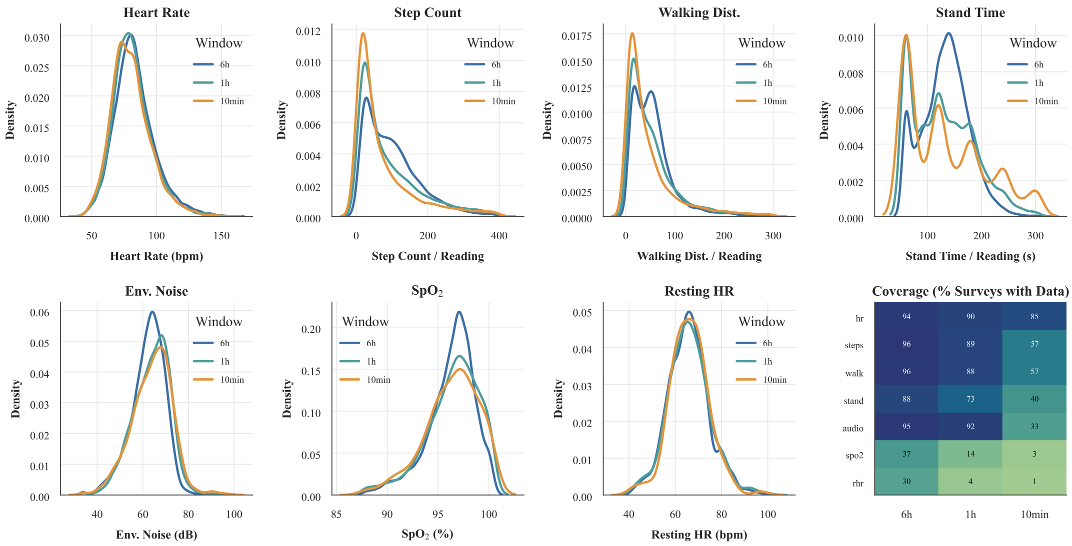

Figure 9. Distributions of pre-survey wearable-health metrics across the 10-minute, 1-hour, and 6-hour aggregation windows: seven wearable health and audio streams as overlaid per-window density curves, plus a metric-by-window data-coverage heatmap.

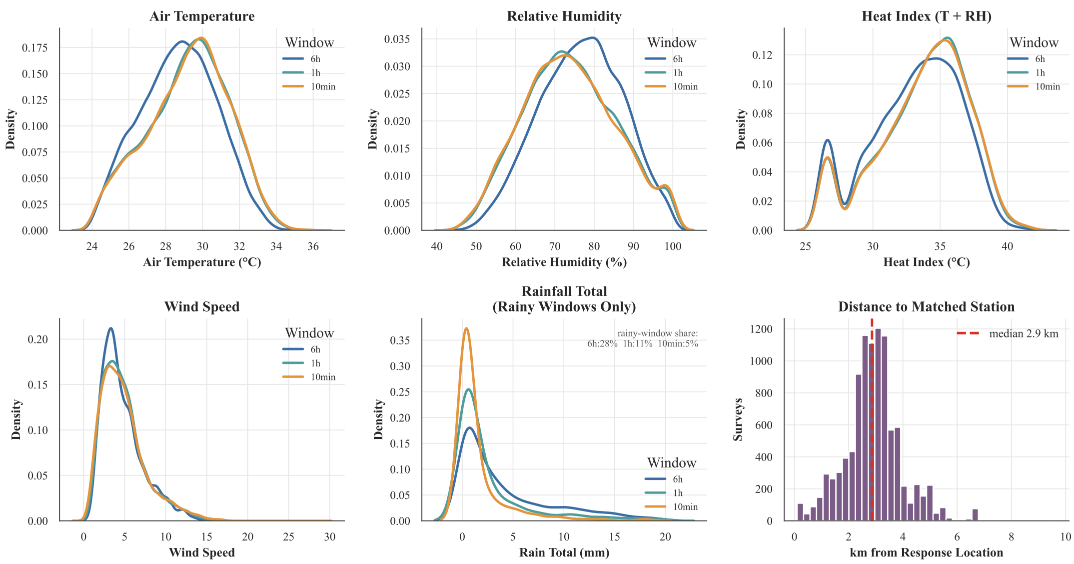

Figure 10. Nearest-station outdoor weather across the 10-minute, 1-hour, and 6-hour pre-survey windows, with distributions for five ambient metrics and the distance to the matched station (median 2.9 km).

3.3. Thermal preference and noise characterization

The two dominant strands of the reported experience, wanting to be cooler and being distracted by noise, were examined before they were related to space type. Cooling preference relative to outdoor heat, across participants, is shown in Figure 11, and the demand is real but only loosely tied to the outdoor thermometer. Across all responses, the share of *Cooler* answers rises with outdoor heat, from about a third when the pre-survey heat index is near 30 °C to roughly half above 37 °C. This situation is expected for a tropical setting, though the outdoor reading is only a proxy for the participant's actual microclimate, which is often a controlled indoor space rather than the outdoor weather. The link between people is weaker than within the heat range. A participant's mean outdoor temperature explains little about who wants cooling ($r \approx 0.33$), and the per-participant panel is very wide, with the *Cooler* share ranging from under 10% for the most heat-tolerant participants to nearly 80% for the most cooling-seeking participants. This spread is a key takeaway of the figure, showing that cooling preference is as much a personal trait as a response to measured conditions, the individual variation that person-level sensing is designed to capture.

The same type of characterization for the sound environment is given in Figure 12, using the linked one-hour sound-level measurement, and here the self-reports and the sensor agree well. Measured sound level rises steadily with perceived noise, from a median near 62 dB for *None* to 66 dB for *A little* and 70 dB for *A lot*. The separation is consistent despite the coarse three-point scale. The measured levels also track where people are, with home and office the quietest settings at around 62 dB while other-indoor, transportation, and outdoor environments sit near 69 to 70 dB, and the sources most often reported as loud, traffic and talking, carry the highest levels. When analyzed together, the two figures show that thermal preference is a personal signal that only loosely tracks the surrounding environment, while noise perception is tied to a measurable exposure that varies with the space around the participant.

3.4. Perceptual fingerprints and association with space type

With each perception characterized on its own, the analysis now asks how the reported experience differs across the spaces in which it was recorded. It does so in three layers that build on one another. The first layer, in Figure 13, reads each space type as a profile by normalizing within its own row. Every cell gives the probability of a response given the space, so each row describes the mix of experiences in that setting rather than how many responses it contributed. Read this way, the spaces separate into recognizable characters. Home is a quiet, solitary setting, with most responses reporting *None* for noise and about 84% reporting *Alone*, while classrooms and other-indoor spaces are mostly social. The commute spaces stand out for discomfort, with the highest shares of *A lot* of noise, reaching roughly one in four responses in transportation against only about 3% at home. Thermal preference shifts more gently, from only about a quarter of office responses wanting *Cooler* to just over half of outdoor responses. This fits the heavily air-conditioned office and the exposed outdoor setting, and it is a reminder that thermal preference reflects the immediate controlled environment more than the outdoor weather. The fingerprint is intuitive but has one limit, since some responses are common everywhere and a high within-space probability does not by itself mean the response is characteristic of that space.

Figure 14 shows the second layer that divides each within-space probability by the response's overall proportion in the dataset. A lift above one marks a response that is

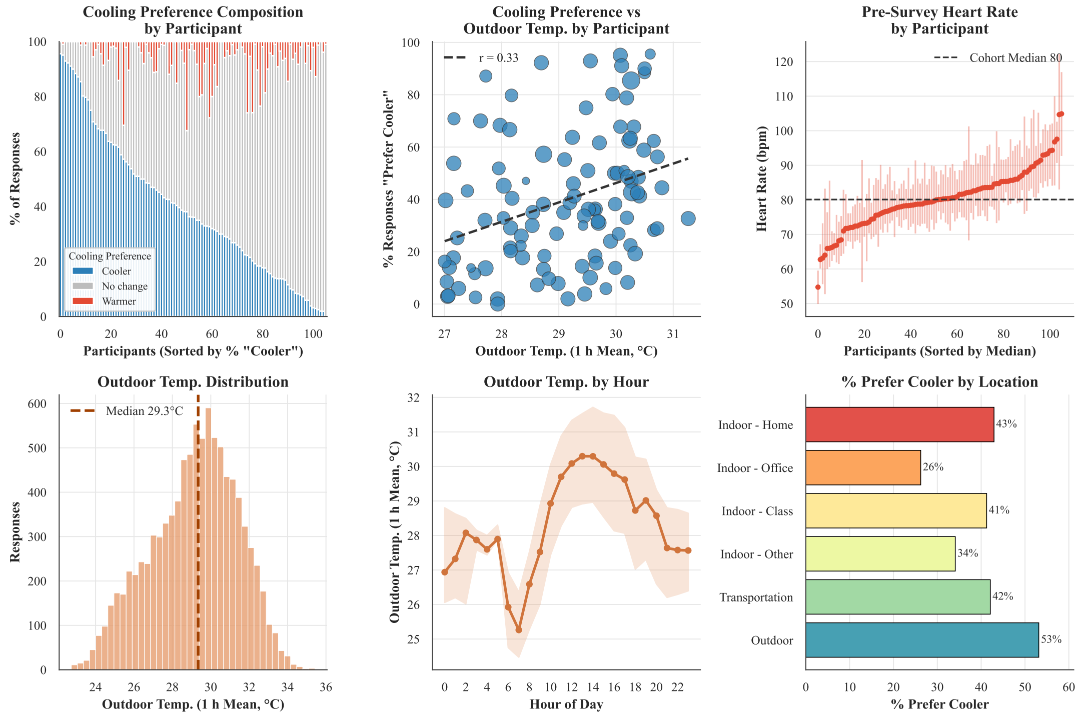

Figure 11. Participant-level and contextual patterns in cooling preference, relating per-participant composition, outdoor temperature, heart rate, and location type.

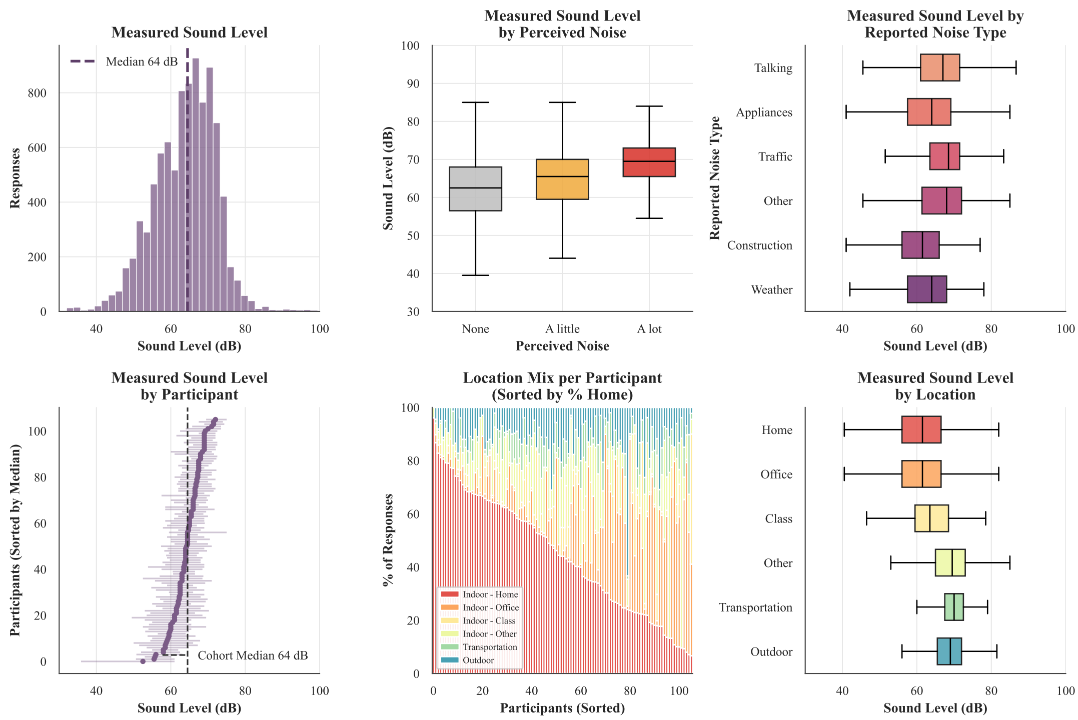

Figure 12. Measured sound level (1-hour pre-survey mean) summarized across subjective, participant, and location contexts.

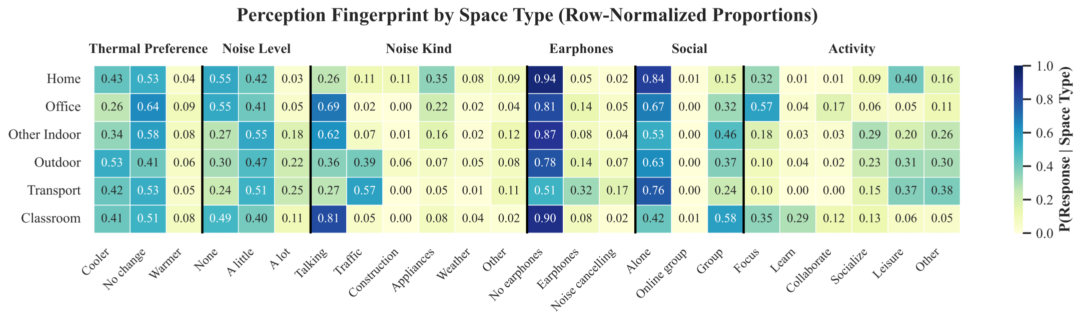

Figure 13. Conditional perception fingerprint by space type in a heatmap of $P(\text{response} \mid \text{space type})$ with space types in rows and response options (grouped by survey dimension) in columns.

more common in a space than in the data as a whole, and a lift below one marks one that is rarer. This turns a profile into a signature, because it isolates what is distinctive to a space rather than what is simply frequent. The contrast sharpens the same story and adds detail the fingerprint hides. *A lot* of noise is over-represented by about 2.6 times in transportation and 2.4 times outdoors, *Group* is over-represented most in classrooms (lift near 2.1), and *Warmer*, a rare response overall, is concentrated in the office (lift near 1.6) where air-conditioning can overshoot. Lift also makes the source of noise a spatial fingerprint in its own right, as traffic accounts for most reported noise in transportation and a large share outdoors, while talking dominates the noise reported in classrooms and offices. The limit of lift mirrors that of the fingerprint, since it flags distinctive combinations but does not show whether any single contrast is larger than chance would produce.

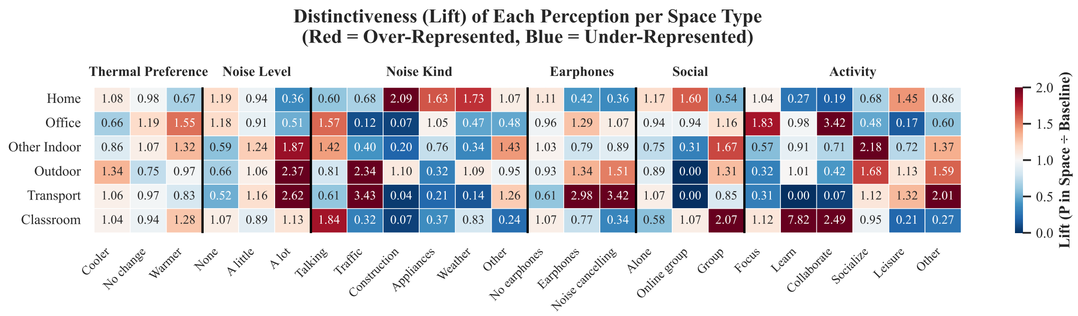

Figure 14. Relative distinctiveness of perceptions by space type, shown as lift, $P(\text{response} \mid \text{space type})/P(\text{response})$, with space types in rows and response options (grouped by survey dimension) in columns.

Figure 15 illustrates the third layer of analysis that adds the statistical dimension the first two lack. It compares each space-response combination against what independence would predict, using adjusted standardized residuals whose sign indicates the direction of the effect and whose magnitude is standardized to a standard-normal scale. This layer confirms which distinctive contrasts are large relative to sampling noise, and it agrees with the overall effect sizes. The association with space type is strongest for what a person is doing and hearing, with Cramér's V near 0.38 for group activity, 0.35 for solitary activity, and 0.28 for the kind of noise, and weakest for thermal preference at about 0.12. The residuals make the behavioral picture explicit, as transportation and outdoor spaces stand out for loud, traffic-driven noise, classrooms and offices for talking and for group or focused activity, and home for quiet, solitary responses. Thermal preference produces only muted residuals scattered across spaces. Because the responses are clustered within 106 participants, the significance markers are read as descriptive screens rather than formal tests, and the agreement of all three layers on the same ordering is the stronger evidence. The lesson across the three layers is that space type is written far more clearly into what people are doing and hearing than into how warm they feel. This weak thermal signal is consistent with people often being in controlled indoor settings rather than being exposed to outdoor weather.

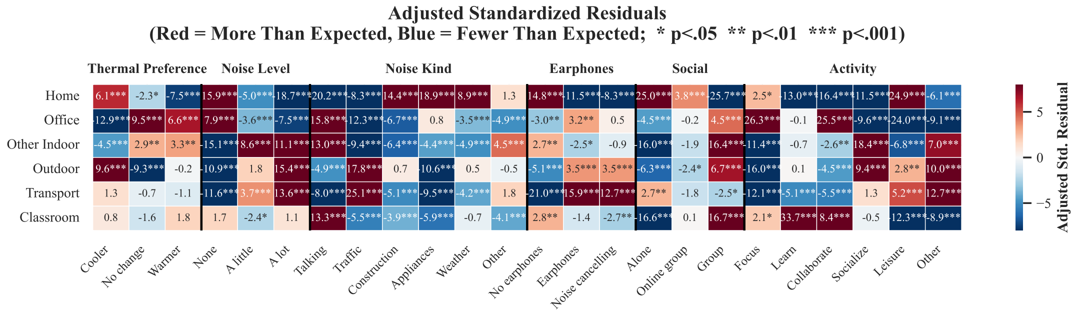

Figure 15. Adjusted standardized residuals for space-type by response combinations, where red cells occur more often and blue cells less often than expected under independence.

4. Discussion

The three analytical layers converge on a coherent characterization of how urban space types differ in reported experience. The clearest separations are behavioral and acoustic rather than thermal, since what people are doing and hearing identifies a space more sharply than how warm they feel. Tables 1 and 2 collect the findings that reinforce prior expectation and establish this baseline character of each setting, grouped by the acoustic environment and by social context and thermal comfort. Each entry pairs a short insight with the supporting detail drawn from the fingerprint, lift, and residual analyses of Section 3.4.

Table 1. Space-type findings that reinforce prior intuition for the acoustic environment.

Commute spaces are the noisiest	Transportation and outdoor responses carry the highest *a lot* noise shares ($\approx$ 25% and 22%), against only about 3% at home.
Home and offices are the quietest settings	Both sit near a 62 dB median sound level, the lowest among the measured spaces.
Perceived loudness tracks the setting	Louder spaces attract more *a lot* noise reports, so subjective loudness follows the physical environment.
Traffic defines commute noise	Traffic is the dominant reported noise source in transportation ($\approx$ 57%) and outdoors ($\approx$ 39%).
Talking defines indoor noise	Talking dominates the acoustic complaints of classrooms ($\approx$ 81%) and offices ($\approx$ 69%).

The findings show that a watch-based micro-survey recovers the structure of human perception in the urban context, some of which aligns with previous studies. Commute spaces carry the highest *a lot* noise shares, and traffic dominates their reported noise

Table 2. Space-type findings that reinforce prior intuition focused on social context and thermal comfort.

Home is overwhelmingly solitary	About 84% of home responses report being *alone*, the most solitary setting measured.
Classrooms are the most social setting	Classrooms are the most group-oriented space, with a group lift near 2.1.
Activity is the strongest spatial marker	Group activity is the response most tightly associated with space type (Cramér's V $\approx$ 0.38).
Offices show the lowest cooling demand	Only about 26% of office responses want *cooler*, consistent with heavy air-conditioning.
Outdoor spaces show the highest cooling demand	Just over half ($\approx$ 53%) of outdoor responses want *cooler*.

source. These results agree with mobile-exposure studies of commuters, where personal exposure to traffic noise and air pollution is context-dependent and is best captured by pairing wearable measurement with reports taken on the move (Marquart, Schlink, & Nagendra, 2022). They also agrees with spatio-temporal urban assessments that place the loudest ambient conditions in traffic-dense cores rather than in residential or green areas (Kumar & Pandey, 2013), and with survey evidence that residential road-traffic noise is a dominant source of annoyance (Preisendörfer, Liebe, Bruderer Enzler, & Diekmann, 2022). That a wrist-based micro-survey, interviews, and remote sensing produce the same ordering of spaces gives more confidence than any single method would, and it supports treating the less obvious findings that follow as reliable.

However, beyond the intuitive elements, several results are less obvious and reshape how the characterization could be read. Thermal preference, the signal one might expect a tropical climate to dominate, is in fact the weakest spatial marker, while distinctiveness that raw proportions understate emerges only once responses are weighed against their overall frequency. Tables 3 and 4 gather these counterintuitive and analytical findings, separating the environmental and thermal surprises from the behavioral signatures and from the framing that effect sizes, rather than p-values, provide under participant nesting.

Table 3. Less intuitive environmental and thermal insights for further exploration.

Thermal preference is the weakest spatial signal	Its association with space type (Cramér's V $\approx$ 0.12) sits far below activity or noise.
Cooling demand separates spaces only modestly	Outdoor demand ($\approx$ 53%) exceeds home ($\approx$ 43%) by less than an air-conditioning-driven intuition would suggest.
Offices can over-cool	Wanting *warmer*, rare overall, concentrates in offices (lift $\approx$ 1.6), a signature of air-conditioning overshoot.
The kind of noise matters more than the amount	The source of noise fingerprints a space more sharply than how loud it is.
Lift exposes hidden distinctiveness	*A lot* of noise is $\approx 2.6\times$ over-represented in transportation relative to the whole corpus, a contrast raw proportions understate.

The fact that activity is the sharpest spatial marker aligns with behavioral accounts of how occupants interact with their environments (Heydarian et al., 2020) and with evidence that movement and exposure are spatially structured within a single setting (Pollard et al., 2022).

Table 4. Method-based and behavioral insights for further exploration.

Behavior separates spaces best	What a person is doing distinguishes space types more sharply than any environmental variable.
Home is distinctively, not just frequently, solitary	Even against an already alone-dominated corpus, home's alone lift exceeds one.
Commuting is loud and lonely	Transportation pairs loud, traffic-driven noise with high solitude ($\approx$ 76% alone), combining discomfort with isolation.
Three methods converge	Fingerprint, lift, and residuals agree on the same behavioral ordering, strengthening confidence beyond any single test.
Effect sizes, not p-values, are the valid read-out	Responses are nested within 106 participants, so significance markers act as descriptive screens rather than formal tests.

4.1. Implications for sustainable urban design

These results help close the loop between how planners assume occupants behave and perceive their surroundings and how occupants actually report experiencing them. Much of the picture reinforces intuition, but the same instrument can flag settings that perform better or worse than expected, and it does so from reports made on site rather than recalled later or judged from images. This matters because how residents read an environment can affect outcomes such as self-rated health more than the measured environment itself (Y. Kim, 2016), so a design process tuned only to physical metrics can miss the perceptions that shape how a space is used. Noise and thermal conditions are consistently among the leading sources of occupant dissatisfaction in large office-survey databases (Frontczak et al., 2012; Parkinson, Schiavon, Kim, & Betti, 2023), so localizing where these complaints arise is directly useful for post-occupancy evaluation. Delivered on a consumer smartwatch, the micro-survey extends the mobile-crowdsourcing approach of collecting geolocated resident feedback (Phuttharak & Loke, 2020) into a continuous channel for post-occupancy evaluation and participatory planning, aligning with the broader agenda of human-building interaction for intelligent built environments (Becerik-Gerber et al., 2022).

From the acoustic perspective, the findings could also support design decisions. Commute and outdoor spaces hold the loudest and most traffic-driven noise, with about a quarter of transportation responses reporting *a lot* of noise against only about 3% at home, so traffic calming, acoustic buffering along transit corridors, and quieter routing would help where distraction is worst. Indoors, talking rather than traffic dominates the noise reported in classrooms and offices, echoing evidence that intelligible background speech is the principal office distraction source (Jahncke, Björkeholm, Marsh, Odelius, & Sörqvist, 2016) and reflecting the trade-off between privacy and communication in open-plan layouts (J. Kim & de Dear, 2013), so the fix there is acoustic treatment and layout that manage speech and reverberation.

The thermal results are related to energy policy and design considerations for a tropical, heavily air-conditioned city. Offices show the lowest cooling demand of any space, with only about a quarter of responses wanting *cooler* and occasional *warmer* requests where cooling overshoots, which suggests that relaxing indoor setpoints could save cooling energy at little cost to comfort. Outdoor spaces show the highest cooling demand, which points to shading, greenery, and other passive cooling of public spaces as the priority for thermal relief. Thermal preference is the weakest spatial marker and reflects the immediate controlled setting more than the outdoor weather, so comfort here is set mainly by the indoor environments people occupy rather than by climate alone.

The layered reading used here fits a broader move toward multi-method assessment of urban perception that combines subjective, physiological, and spatial data, as in work that joins participatory mapping, neurophysiological response, and self-reported emotion to study perceived walkability (Yıldırım, Heyik, Güleç Ozer, & Sungur, 2026). Because the same watch-based platform has already supported city-scale prediction of noise distraction and thermal preference (Miller, Ibrahim, et al., 2025), the structure described here could feed predictive, just-in-time design tools embedded in urban digital twins (Lei et al., 2026) and the urban-scale occupant-behavior models used to anticipate building energy demand (Salim et al., 2020), tools that anticipate where and when discomfort arises. Heat and noise are only two of the dimensions such an instrument can capture, and the same loop could be closed for others.

4.2. Limitations

Generalizability is a limitation in this analysis as the findings come from 106 participants in a young and homogeneous cohort, with roughly 73% aged 18 to 30 and most recruited as university students. Recruitment was by convenience rather than through a representative sampling frame, so the reported patterns describe this cohort in Singapore and should not be read as population-level estimates. The setting itself also limits transfer, since Singapore is a dense, tropical, heavily air-conditioned city, and the thermal results in particular reflect that context more than a general climate response.

The structure of the data, rather than a controlled design, also shapes what the analysis can claim. Although there are 9,962 responses, they are nested within only 106 participants, so responses from the same person are not independent. We therefore treat the associations and significance markers as descriptive screens rather than formal tests, and we do not fit a model that accounts for this clustering. Perceptions are self-reported through single-item micro-surveys answered in the moment, and the reported space type is a participant label rather than a verified location, so both carry the usual subjectivity and recall effects of momentary self-report.

Uneven coverage across the dataset qualifies the finer comparisons, beginning with the conditional follow-up questions such as noise kind, activity, and earphone use, which are answered only by the responses that reach them and so rest on smaller samples than the always-asked items. Within-space cells built from a few responses are noisier than their shading suggests, because no smoothing or shrinkage is applied. Linked streams are also uneven, as some physiological signals, such as blood oxygen and resting heart rate, are sparse, and the early-morning hours are thinly sampled, with under 2% of responses before 08:00, so temporal patterns in those hours are weak. The space types themselves are unbalanced, with nearly half of all responses coming from home, so the corpus-wide baselines used for lift are weighted toward the most common settings.

Exposure and scope raise a final pair of limitations, beginning with the weather values, which are aggregated from the nearest station and describe ambient outdoor conditions rather than the microclimate or skin-level exposure each participant actually experienced, a coarse proxy especially for the indoor and air-conditioned settings where many responses were made. Finally, the deployment ran in two phases that differed in how just-in-time messages were triggered, and the present analysis pools across them, so it characterizes reported experience and does not isolate any effect the intervention messages may have had on what participants reported or did.

4.3. Future work

The most direct next step is to broaden the cohort through larger deployments that run fully on participants' own smartwatches with remote self-onboarding, which could reach thousands or tens of thousands of people and a wider mix of ages, occupations, and living situations. A more diverse cohort would allow the reported patterns to be tested for generalizability rather than described for one group, and replication in other cities would show which findings are specific to a dense, tropical, air-conditioned setting and which hold more broadly.

Future analyses could also treat the nested structure of the data directly, using mixed-effects or otherwise participant-grouped models that account for repeated responses from the same person, turning the descriptive associations reported here into estimates with proper uncertainty and supporting participant-grouped validation for any predictive work built on the same data. Exposure measurement could improve by pairing the micro-surveys with personal or wearable microclimate and air-quality sensing, such as on-body temperature, humidity, and sound level, where guidance on low-cost sensors is now maturing (Parkinson et al., 2026), which would replace the coarse nearest-station weather with the conditions each participant actually experienced and would matter most in the indoor and air-conditioned settings that dominate the responses.

The survey design itself could become more targeted, with prompts that adapt to where a person is and to the specific subjective question a deployment is meant to answer, concentrating responses where they are most informative and filling the thinly sampled settings and hours noted above. Building on this, a deeper characterization of participants who share behavioral patterns could establish behavioral archetypes, helping identify which patterns to encourage through design and which to discourage before they become common. In addition, longer and more structured deployments would extend the scope, since running over months rather than weeks would capture seasonal and longer-term variation, and designs that separate or randomize the just-in-time intervention phases would let future work isolate the effect of the messages on reported experience and behavior, which the present pooled analysis cannot.

5. Conclusion

This paper characterizes the everyday human experience of urban space types in a dense tropical city using scalable, crowdsourced smartwatch micro-surveys. A field deployment of the open-source Cozie platform in Singapore collected 9,962 geolocated responses from 106 participants, each linking self-reported thermal preference, noise, social context, and activity to onboard physiological and linked weather data. The reported experience was characterized across six everyday space types through conditional perception fingerprints, relative distinctiveness, and effect-size-based association testing. The results outline that what people do and hear separates urban spaces more clearly than how warm they feel. Activity and the kind of noise are the strongest spatial markers, while thermal preference is the weakest, reflecting the heavily air-conditioned indoor settings that dominate daily life more than the outdoor climate. Commute spaces emerge as the loudest and most traffic-driven, indoor spaces are defined by talking, and cooling demand is lowest in offices and highest outdoors. Much of this reinforces intuition, but the value of the approach is that it recovers and quantifies these signatures directly from lived, in-situ reports rather than from assumptions built into design. The method also has clear limits, from a young and homogeneous cohort to responses nested within participants and ambient rather than personal exposure, each of which points to a direction for future work. Taken together, the results show that a wrist-based micro-survey can provide a human-centered evidence base for noise mitigation and for energy-conscious, thermal-comfort-aware urban design, and they set the stage for larger and more diverse deployments across cities.

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CRediT authorship contribution statement

Clayton Miller: Conceptualization, Methodology, Software, Formal analysis, Resources, Writing – original draft, Visualization, Supervision, Project administration, Funding acquisition. Mario Frei: Conceptualization, Methodology, Software, Validation, Investigation, Data curation, Writing – review & editing, Project administration. Yun Xuan Chua: Conceptualization, Methodology, Validation, Investigation, Writing – review & editing, Project administration.

Declaration of generative AI use

During the preparation of this work, the authors used Runcell AI (Version 0.1.25), an AI coding and writing assistant built on the GPT-5.6 Terra (OpenAI) and Claude Opus 4.8 (Anthropic) large language models, to help organize and format the LaTeX manuscript, arrange figure and table placement, and copy-edit text written by the authors. The tools were not used to generate or analyze the study data, nor to create or alter any figure, its underlying values, or the reported results. All AI-assisted output was reviewed, edited, and verified by the authors, who take full responsibility for the originality, accuracy, integrity, citations, and final content of this manuscript.

Disclosure statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability statement

The data, code, and documentation supporting the findings of this study are available on GitHub at `https://github.com/buds-lab/cozie-singapore-journal`.

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